# Report on

**“Tutorials”**

For

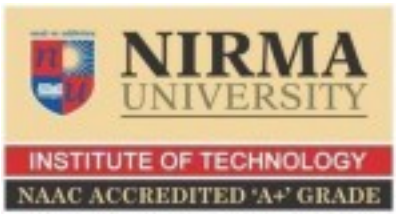
**Software Project Management - 4CS504ME25 B.**

**Tech. Semester IV**

## Prepared By :

Gandhi Urva Yogeshkumar (23BCE078)

Gajnotar Rakshit (23BCE077)




**Institute of Technology,**
**Nirma University**
**Ahmedabad - 382481**

**September 2026**

## Aim

To analyze process and project metrics, compute size-oriented and function-oriented metrics and evaluate quality metrics (Defect Removal Efficiency, Reliability and Availability) for the Hospital Management System (HMS).

## Objective

- Set out a project baseline scenario for the 16-week, 8-sprint HMS lifecycle.
- Calculate size-oriented metrics: Errors/KLOC, Defects/KLOC, Cost/LOC, Documentation/KLOC and Productivity (LOC/PM).
- Estimate code size distribution between backend (Urva) and frontend (Rakshit).
- Compute Function Points using five information domains and 14 General System Characteristics (GSCs).
- Evaluate function-oriented metrics: Cost/FP, FP/PM, Errors/FP, Defects/FP and Documentation/FP.
- Assess quality metrics including Correctness, Defect Removal Efficiency (DRE) and system Reliability and Availability (MTTF, MTTR, MTBF and Availability %).
- Put together all metrics into a summary dashboard to find process improvements.

## 1. Introduction

A metric is a number that measures something about software or how it is made. Without measurement, a manager can only guess whether a project is on schedule or within budget. Metrics give concrete numbers so we can spot trouble early and improve our work.

In software engineering, metrics are divided into three groups:

- Process Metrics: Measure the development process, such as defect removal during testing or velocity across sprints.
- Project Metrics: Track project health, including schedule variance, effort, cost and team productivity.
- Product Metrics: Measure the software itself, such as code size, complexity and system uptime.

In this tutorial we continue with the Hospital Management System (HMS) and calculate three families of metrics: size-oriented metrics based on lines of code (LOC), function-oriented metrics based on Function Points (FP) and quality metrics covering defect removal and system reliability.

## 2. Project Scenario and Baseline Data

The scenario below describes the first release of the Hospital Management System built by Urva (backend) and Rakshit (frontend) over the 16-week semester. The table below lists the baseline values used for our calculations:

| Project Parameter | Baseline Value | Basis / Derivation in HMS Context |
|---|---|---|
| Delivered Code Size | 3,300 LOC (3.3 KLOC) | Estimated across 6 core functional modules and auth/setup (Table 2). |
| Total Effort Expended | 8.0 Person-Months (PM) | 2 developers * 4 calendar months (16 weeks, 8 sprints). Total 1,280 hours. |
| Total Project Cost | ₹1,80,000 | Labor: 2 * ₹20,000/mo * 4 mo = ₹1,60,000 + ₹20,000 lab and machine overhead. |
| Documentation Produced | 120 Pages | SRS (35p), Architecture and DB Schema (30p), |

| Project Parameter | Baseline Value | Basis / Derivation in HMS Context |
|---|---|---|
| | | API specs (25p) and User Guide (30p). |
| Pre-Release Errors Caught | 48 Errors | Found through peer code reviews, unit tests and sprint validation. |
| Post-Release Defects | 5 Defects | Reported by hospital staff during the 3-month pilot trial. |
| Function Points Delivered | 164 FP | Calculated from five information domains and 14 GSCs (Section 4). |
| Observed Operating Period | 3 Months (2,190 Hours) | Continuous pilot trial period observed after deployment. |
| System Failures in Period | 4 Failures | Service disruptions recorded during the 3-month trial. |
| Total Repair Downtime | 6.0 Hours | Total time taken to diagnose, fix and redeploy. |

## 3. Size-Oriented Metrics

Size-oriented metrics are found by dividing measured outcomes by code size in thousands of lines of code (KLOC). The code size across our HMS modules is shown in the table below:

| Module ID | Deliverable Name | Backend LOC (Urva) | Frontend LOC (Rakshit) | Total LOC |
|---|---|---|---|---|
| M1 | Patient Registration and Management | 260 | 280 | 540 |
| M2 | Doctor and Department Management | 220 | 240 | 460 |
| M3 | Appointment Booking and Scheduling | 280 | 300 | 580 |
| M4 | Pharmacy Stock Module | 240 | 260 | 500 |
| M5 | Billing and Invoice Module | 260 | 250 | 510 |
| M6 | Admin Dashboard and Reports | 200 | 270 | 470 |
| M0 | Setup, Auth and Common Helpers | 140 | 100 | 240 |
| **Total** | **Complete System Size** | **1,600** | **1,700** | **3,300** |

Using these numbers, we calculate the size-oriented metrics below:

| Metric | Formula | Calculation | Result |
|---|---|---|---|
| Errors per KLOC | Errors / KLOC | 48 / 3.3 | 14.55 Errors / KLOC |
| Defects per KLOC | Defects / KLOC | 5 / 3.3 | 1.52 Defects / KLOC |
| Cost per LOC | Total Cost / LOC | ₹1,80,000 / 3,300 | ₹54.55 per LOC |
| Cost per KLOC | Total Cost / KLOC | ₹1,80,000 / 3.3 | ₹54,545 per KLOC |
| Documentation per KLOC | Pages / KLOC | 120 / 3.3 | 36.36 Pages / KLOC |
| Developer Productivity | LOC / Effort (PM) | 3,300 / 8.0 | 412.5 LOC / PM |

### Analysis of Size-Oriented Results

- Defect Density: 14.55 errors per KLOC were caught during testing, while only 1.52 defects per KLOC escaped to the users. This shows that most problems were resolved before release.
- Productivity: An output of 412.5 LOC per person-month works out to about 25.8 lines of tested code per person each week. For a modern stack with Node.js and React, this rate is sensible because packages handle basic tasks while we write custom business rules.
- Cost per LOC: At ₹54.55 per line (₹54,545 per KLOC), project costs stayed within our college budget.
- Documentation: 36.36 pages per KLOC reflects the detailed reports required for university coursework, including SRS, ER models, API routes and test logs.

## 4. Function-Oriented Metrics

Function-oriented metrics measure software size by user functionality instead of code lines. Function Points (FP) do not depend on the programming language or framework used.

We counted the functional components across five standard information domains:

- External Inputs (10): Patient registration form, doctor profile entry, appointment booking, prescription entry, medicine stock update, billing entry, payment confirmation, login auth, department assignment and patient status updates.
- External Outputs (7): Patient ID card slip, appointment token, prescription summary, itemized billing invoice, daily revenue report, doctor schedule list and low-stock alerts.
- External Inquiries (8): Search patient by ID, check doctor availability, check appointment status, search drug inventory, check invoice status, view medical history, list doctor appointments and view admin KPI summary.
- Internal Logical Files (5): Patient records store, doctor directory, appointment ledger, pharmacy stock catalog and billing store.
- External Interface Files (1): Payment gateway integration interface for online patient billing.

Using average complexity weights, the Unadjusted Function Points (UFP) total 164:

| Information Domain | Count | Complexity Weight | Weighted FP |
|---|---|---|---|
| External Inputs (EI) | 10 | 4 (Average) | 40 |
| External Outputs (EO) | 7 | 5 (Average) | 35 |
| External Inquiries (EQ) | 8 | 4 (Average) | 32 |
| Internal Logical Files (ILF) | 5 | 10 (Average) | 50 |
| External Interface Files (EIF) | 1 | 7 (Average) | 7 |
| **Total Unadjusted Function Points (UFP)** | **31** | **-** | **164** |

### Value Adjustment Factor (VAF) and Final Function Points

The 14 General System Characteristics (GSCs) cover data communications, performance, transaction rates, end-user efficiency and online updates. In our academic project, each factor is scored from 0 to 5. The total sum is 35:

VAF = 0.65 + (0.01 * 35) = 1.00

Adjusted Function Points = UFP * VAF = 164 * 1.00 = 164.0 FP

The function-oriented metrics from this calculaton are listed below:

| Metric | Formula | Calculation | Result |
|---|---|---|---|
| Cost per FP | Total Cost / FP | ₹1,80,000 / 164 | ₹1,097.56 per FP |
| Productivity (FP) | FP / Effort (PM) | 164 / 8.0 | 20.50 FP / PM |
| Errors per FP | Errors / FP | 48 / 164 | 0.29 Errors / FP |
| Defects per FP | Defects / FP | 5 / 164 | 0.030 Defects / FP |
| Documentation per FP | Pages / FP | 120 / 164 | 0.73 Pages / FP |

### Analysis of Function-Oriented Results

- Delivery Rate: The team delivered 20.50 FP per person-month, which means about 7.8 hours per function point (1,280 hours / 164 FP). This matches typical web project benchmarks.
- Cost per FP: At ₹1,098 per function point, delivery stayed economical under our student stipend budget.
- Fair Comparison: Unlike LOC, the 164 FP figure gives an objective measurment that can be compared against later releases even if we rewrite parts in another framework.

## 5. Quality Metrics

Quality metrics tell us how well the delivered software works and whether users run into bugs during regular hospital tasks.

### 5.1 Correctness

Correctness is measured by counting defects reported by users per KLOC. In our 3-month trial, 5 defects were found in 3.3 KLOC, giving 1.52 defects per KLOC.

### 5.2 Defect Removal Efficiency (DRE)

DRE measures how good the team is at catching bugs before the software reaches users:

DRE = [ E / (E + D) ] * 100

Here E is the 48 errors caught during development and testing and D is the 5 defects reported by users:

DRE = [ 48 / (48 + 5) ] * 100 = (48 / 53) * 100 = 90.57%

A DRE above 90% is the target for well-run projects. Our result of 90.57% shows that testing in week 2 of each sprint caught nine out of ten bugs before release.

### 5.3 Reliability, Maintainability and Availability

During the 3-month (2,190 hour) trial period, 4 failures occured, causing 6.0 hours of total repair downtime:

| Metric | Formula | Calculation | Result |
|---|---|---|---|
| Correctness | Defects / KLOC | 5 / 3.3 | 1.52 Defects / KLOC |
| Defect Removal Efficiency | E / (E + D) * 100 | 48 / (48 + 5) * 100 | 90.57% |
| Mean Time To Failure (MTTF) | (Operating Hrs - Downtime) / Failures | (2,190 - 6) / 4 | 546.0 Hours |

| Metric | Formula | Calculation | Result |
|---|---|---|---|
| Mean Time To Repair (MTTR) | Total Downtime / Failures | 6.0 / 4 | 1.50 Hours |
| Mean Time Between Failures | MTTF + MTTR | 546.0 + 1.5 | 547.5 Hours |
| Failure Rate (λ) | 1 / MTTF | 1 / 546.0 | 0.00183 Failures / Hr |
| Operational Availability | MTTF / (MTTF + MTTR) * 100 | 546.0 / 547.5 * 100 | 99.73% |

- Reliability: An MTTF of 546.0 hours (nearly 23 days between failures) shows steady day-to-day operation.
- Maintainability: An MTTR of 1.50 hours indicates that our modular Express routes and React components make it straightforward to locate and patch problems.
- Availability: An availability of 99.73% means the system was ready for hospital reception and doctors with very little downtime.

## 6. Summary and Process Improvement

The table below brings together the size-oriented, function-oriented and quality metrics calculated for the Hospital Management System:

| Metric Category | Metric Name | Calculated Value | Interpretation |
|---|---|---|---|
| Size-Oriented | Errors per KLOC | 14.55 | Internal testing found bugs early in sprint cycles. |
| Size-Oriented | Defects per KLOC | 1.52 | Low defect escape rate in pilot usage. |
| Size-Oriented | Cost per LOC | ₹54.55 | Economical development within student budget. |
| Size-Oriented | Developer Productivity | 412.5 LOC/PM | Consistent speed for Node.js and React. |
| Function-Oriented | Cost per FP | ₹1,098 | Low cost per delivered feature. |
| Function-Oriented | FP Productivity | 20.50 FP/PM | Steady feature delivery across 8 sprints. |
| Quality | Defect Removal Efficiency | 90.57% | Reaches the 90% benchmark. |
| Quality | MTTF / MTTR / MTBF | 546.0h / 1.5h / 547.5h | Steady operation and quick bug fixes. |
| Quality | Operational Availability | 99.73% | High uptime suitable for clinic tasks. |

### Actionable Process Observations

- Strengths: Defect Removal Efficiency (90.57%) and availability (99.73%) show that giving week 2 of each sprint to testing kept user bugs low.
- Weaknesses: Documentation effort was high (36 pages/KLOC) because of manual report writing. Also, pharmacy stock and billing edge cases caused 3 of the 5 defects.

- Process Recommendations: For next sprints, we will add automated API tests with Supertest and Jest to check billing calculations automatically before final integration.

## 7. Conclusion

This report set up a complete measurement framework for the Hospital Management System. Size-oriented metrics showed project scale (3.3 KLOC and ₹54.55/LOC) and team productivity (412.5 LOC/PM). Function-oriented metrics gave a language-neutral measure (164 FP at ₹1,098/FP). Quality metrics verified good engineering with 90.57% Defect Removal Efficiency and 99.73% availability. These figures provide clear baseline numbers for the estimation in T7.